

4.1 Forces: Turning Effects & Equilibrium

Question Paper

Course	CIEA Level Physics
Section	4. Forces, Density & Pressure
Topic	4.1 Forces: Turning Effects & Equilibrium
Difficulty	Hard

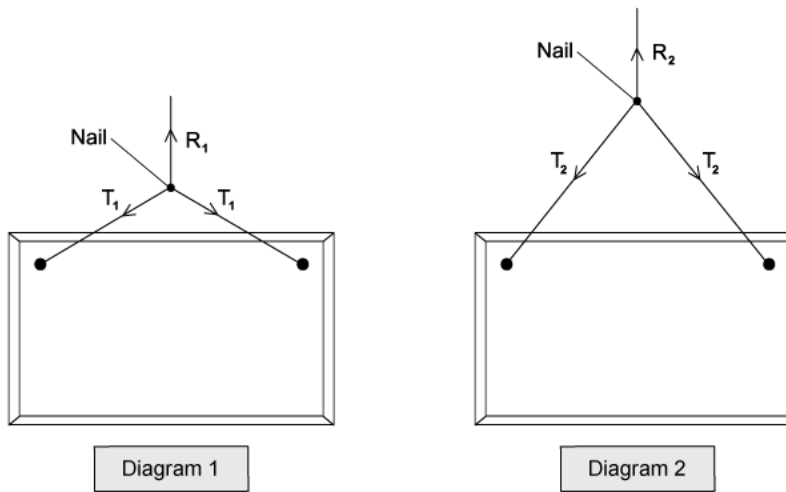
Time allowed: 20

Score: /10

Percentage: /100

Question 1

The diagrams show two ways of hanging the same picture.



In both cases, a string is attached to the same points on the picture and looped symmetrically over a nail in a wall. The forces shown are those that act on the nail.

In diagram 1, the string loop is shorter than in diagram 2.

Which information about the magnitude of the forces is correct?

A. $R_1 = R_2$ $T_1 < T_2$

B. $R_1 = R_2$ $T_1 > T_2$

C. $R_1 > R_2$ $T_1 < T_2$

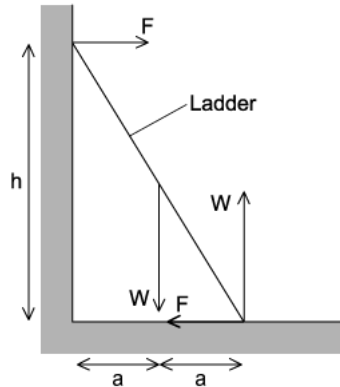
D. $R_1 < R_2$ $T_1 = T_2$

[1 mark]

Question 2

A uniform ladder rests against a vertical wall where there is negligible friction. The bottom of the ladder rests on rough ground where there is friction. The top of the ladder is at a height h above the ground and the foot of the ladder is at a distance $2a$ from the wall.

The diagram shows the forces that act on the ladder.



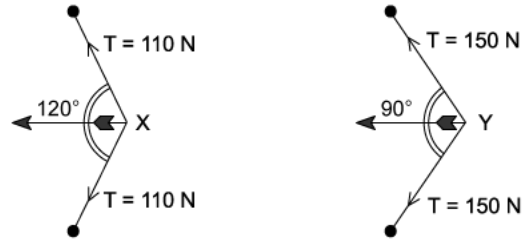
Which equation is formed by taking moments?

- A. $Wa + Fh = 2Wa$
- B. $Fa + Wa = Fh$
- C. $Wa + 2Wa = Fh$
- D. $Wa - 2Wa = 2Fh$

[1 mark]

Question 3

An archer draws his bowstring back to position X. The bowstring and arrow are shown. The tension T in the string is also shown. Then he draws the bowstring back further to position Y.



The archer finds the resultant force on the arrow is greater when the arrow is released from position Y.

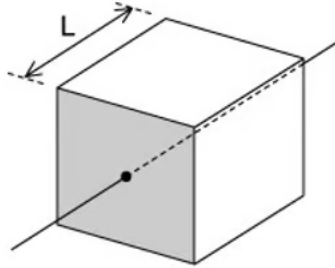
What is the increase in force?

- A. 51 N
- B. 55 N
- C. 102 N
- D. 110 N

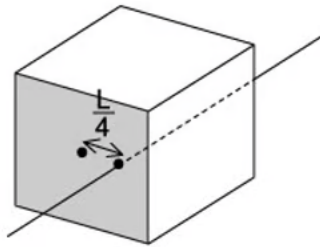
[1 mark]

Question 4

The diagram shows a solid cube with weight W and sides of length L . It is supported by a frictionless spindle that passes through the centres of two opposite vertical faces. One of these faces is shaded.



The spindle is now removed and replaced at a distance $\frac{L}{4}$ to the right of its original position.



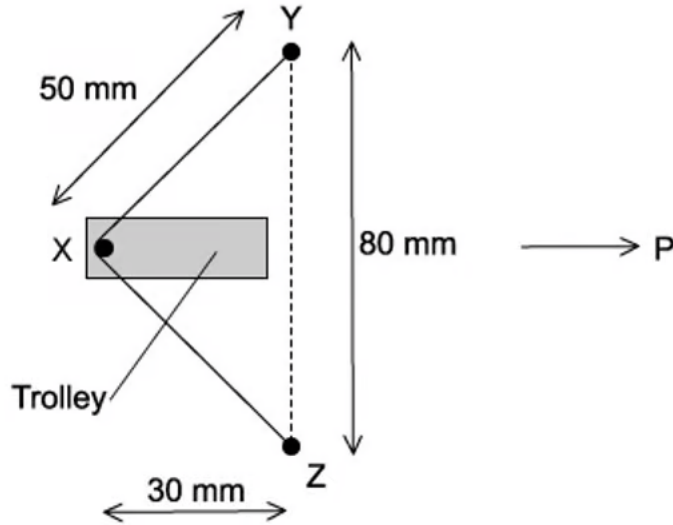
When viewing the shaded face, what is the torque of the couple that will now be needed to stop the cube from toppling?

- A. $\frac{WL}{2}$ anticlockwise
- B. $\frac{WL}{2}$ clockwise
- C. $\frac{WL}{4}$ anticlockwise
- D. $\frac{WL}{4}$ clockwise

[1 mark]

Question 5

The diagram shows two fixed pins, Y and Z. A length of elastic is stretched between Y and Z and around pin X, which is attached to a trolley.



X is at the centre of the elastic and the trolley is to be propelled in the direction P at right angles to YZ. The tension in the elastic is 4 N.

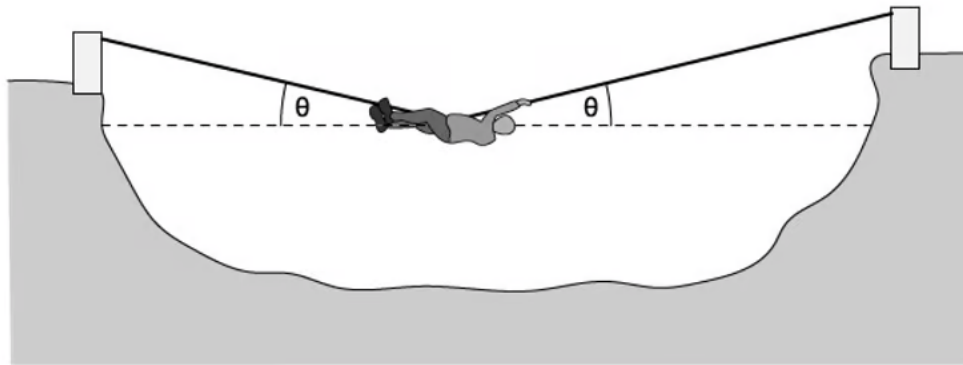
What is the force accelerating the trolley in the direction P when the trolley is released?

- A. 2.4 N
- B. 3.2 N
- C. 4.8 N
- D. 6.4 N

[1 mark]

Question 6

The diagram shows a rope bridge that a student makes on an adventure training course. The student has a weight W .



Which formula gives the tension T in the rope?

A. $T = \frac{W}{2 \cos \theta}$

B. $T = \frac{W}{2 \sin \theta}$

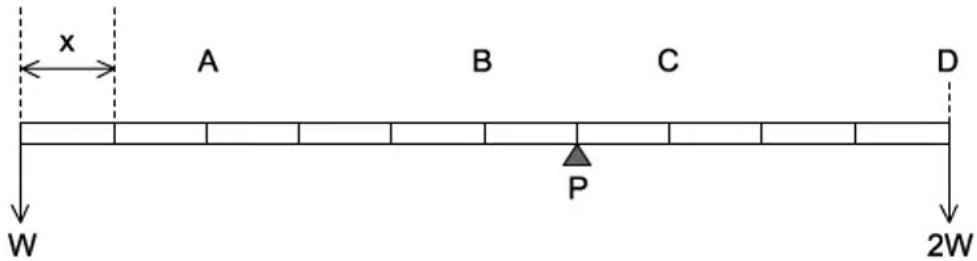
C. $T = \frac{W}{\cos \theta}$

D. $T = \frac{W}{\sin \theta}$

[1 mark]

Question 7

A uniform beam is pivoted at P as shown. Weights of W and $2W$ are attached to its ends. The beam is marked at x intervals. The weight of the beam is $10W$

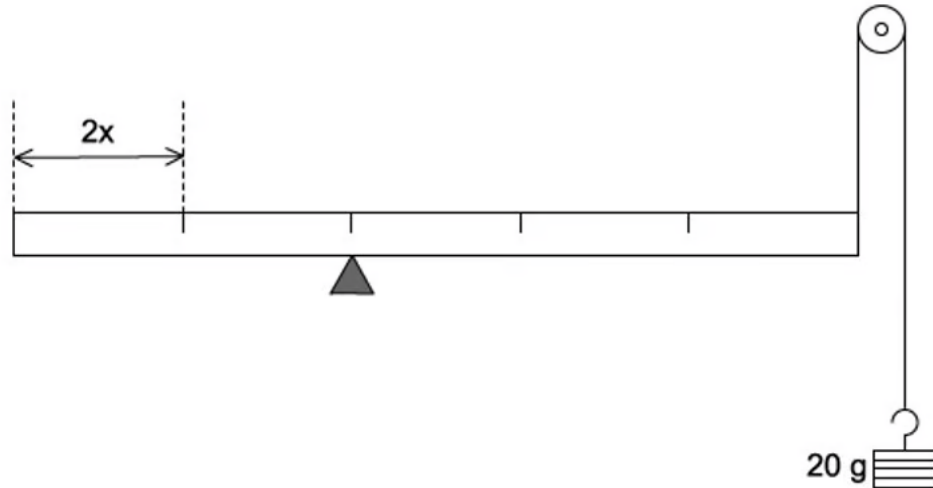


At which point should a $2W$ weight be attached to achieve equilibrium?

[1 mark]

Question 8

A uniform metre rule of mass 100 g is supported by a pivot and a string as shown in the diagram. The beam is marked at $2x$ intervals. The string passes round a frictionless pulley and carries a mass of 20 g.



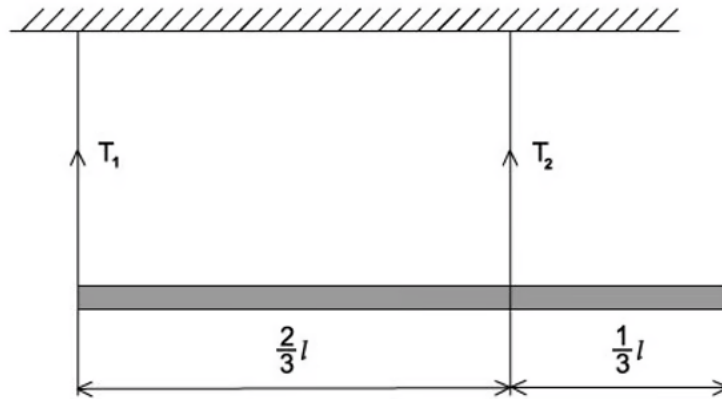
At which mark on the rule must a 50 g mass be suspended so that the rule balances?

- A. $0.4x$
- B. $3.6x$
- C. $4.4x$
- D. $6.4x$

[1 mark]

Question 9

A heavy uniform beam of length L is supported by two vertical cords as shown



What is the ratio of $\frac{T_1}{T_2}$ of the tensions in these cords?

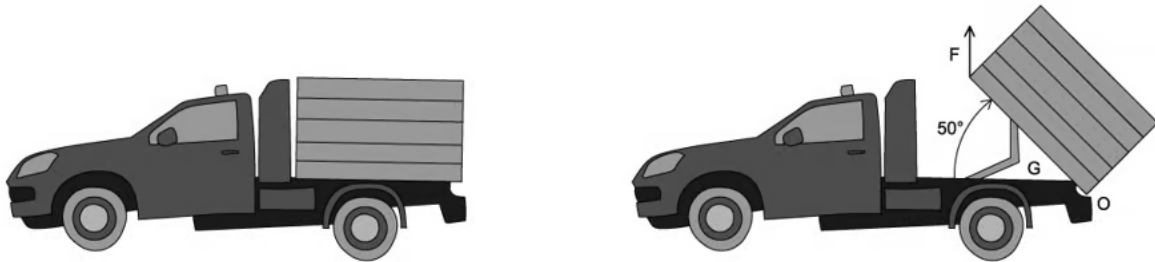
- A. $\frac{1}{3}$
- B. $\frac{1}{2}$
- C. $\frac{2}{1}$
- D. $\frac{3}{1}$

[1 mark]

Question 10

A trailer of weight 30 kN is attached to a truck. The trailer is raised until it is held at 50° to the horizontal by a vertical force F as shown in the diagram.

The trailer is 2.7 m in length and the position of its centre of gravity G is a third of the length from the hinge at O .



What is the vertical force F applied to the end of the trailer?

- A. 10 kN
- B. 15 kN
- C. 30 kN
- D. 60 kN

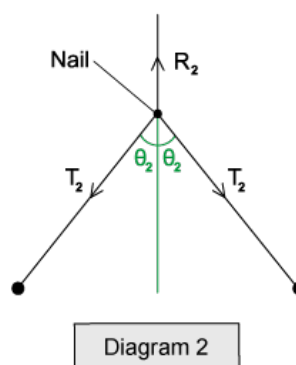
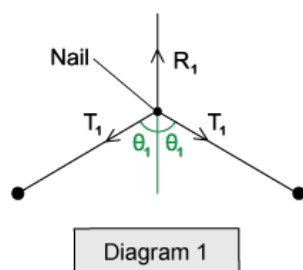
[1 mark]

Model Answers: Hard

1

The correct answer is **B** because:

- The weights of both pictures are the same, therefore, the upward tensions in both strings are the same, therefore:
 - $R_1 = R_2$
- To determine which is the larger of T_1 or T_2 , we must consider the angles between the strings and the vertical, let's call these θ_1 and θ_2
- For each diagram, the sum of the vertical downward components of the tensions should be equal to the weight which is also equal to R
 - $W = 2T_1 \cos \theta_1$
 - $W = 2T_2 \cos \theta_2$
 - $2T_1 \cos \theta_1 = 2T_2 \cos \theta_2$
- It is clear from the diagrams that the angle θ_1 is larger than θ_2
- From the graph of cosine, we know that as the angle θ increases, the value of $\cos \theta$ decreases
- Since $\theta_1 > \theta_2$, $\cos \theta_1$ is smaller than $\cos \theta_2$, so for the statement to hold true, we can conclude:
 - $T_1 > T_2$



A is incorrect because this implies that $\theta_1 < \theta_2$, which is incorrect

C & D are incorrect because R is equal to the weight of the picture so both of these are incorrect

2

The correct answer is **A** because:

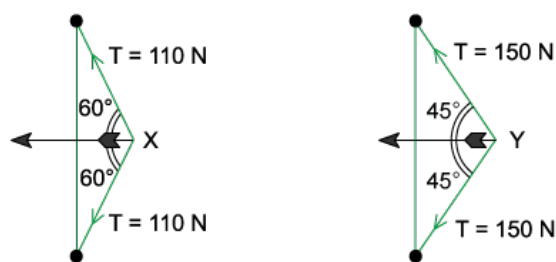
- The ladder is in equilibrium since it is not moving
- The moment of a force is equal to
 - *Moments = Force $\times$ perpendicular distance from pivot*
- The most important part of this question to find the most appropriate point on the ladder to take moments
- We can consider the pivot around either the top or bottom of the ladder, either option will give the correct result, but we will take the moments about the bottom of the ladder in this case
- **Clockwise moments:**
- The force F at the top of the ladder acts at a height h , so, this causes a clockwise moment = Fh
- The weight W of the ladder acts at a perpendicular distance of a causing a clockwise moment of = Wa
- **Anticlockwise moments:**
- The ground causes a reaction force W upwards on the ladder at a distance of $2a$ from the bottom corner. This causes an anticlockwise moment of = $2Wa$
- The force F at the bottom of the ladder is not perpendicular to the force so no moment is produced here
- For equilibrium: Sum of clockwise moments = Sum of anticlockwise moments
 - $Wa + Fh = 2Wa$

3

The correct answer is **C** because:

- Hopefully you will have quickly guessed that this problem will require trigonometry to solve
- In both cases, the bowstring is in equilibrium, so we can work out the resultant force in each bow using:
 - $F = T \cos \frac{\theta}{2} + T \cos \frac{\theta}{2} = 2T \cos \frac{\theta}{2}$
- Where T is the tension in the string and θ is the angle between the strings

- The $2T$ is because there is tension on both halves of the string and the $\frac{1}{2}\theta$ is because you resolve in line with the direction that the archer is pulling, which is half the angle between the halves of the bow
 - Resultant force in X = $2T \cos \frac{\theta}{2} = 2 \times 110 \cos \frac{120}{2} = 220 \cos 60$
 - Resultant force in Y = $2T \cos \frac{\theta}{2} = 2 \times 150 \cos \frac{90}{2} = 300 \cos 45$
 - Increase in force = $300 \cos 45 - 220 \cos 60 = 102 \text{ N}$



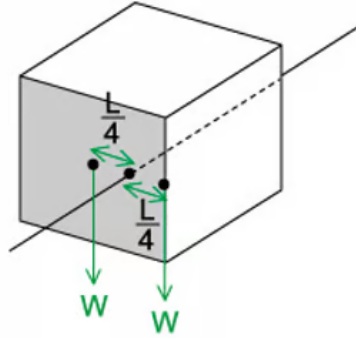
Exam Tip: if there is more than one force triangle in a question, always consider each one separately, many students consider the two strings as one and end up with the wrong answer, so be careful!

4

The correct answer is **D** because:

- We need to find the torque required to maintain equilibrium. This is not the torque produced by the centre of mass of the cube
- The weight acts downwards and is on the left on the pivot, therefore, this causes an anticlockwise moment at a perpendicular distance of $\frac{L}{4}$ from the spindle
 - Anticlockwise moment due to weight = $W \times \frac{L}{4} = \frac{WL}{4}$
- For equilibrium: Sum of clockwise moments = Sum of anticlockwise moments
- The question asks for the torque of the couple. In a couple, the distance between the forces, the one causing the clockwise moment and the one causing the anticlockwise moment, should be the same distance from which the weight acts
- So, to maintain equilibrium, a force equal to W should act along the vertical edge of the shaded face, which is a distance of $\frac{L}{4}$ to the right of the pivot

- Clockwise moment to be applied $= W \times \frac{L}{4} = \frac{WL}{4}$
- Therefore, the torque of the couple that is required to prevent toppling is $\frac{WL}{4}$ clockwise



5

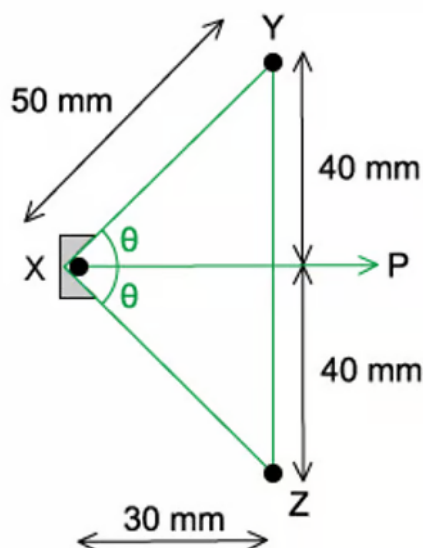
The correct answer is **C** because:

- If we set up force triangles, we can see the tension in each string is equal to
 - $T \cos \theta$
 - Where T is the tension in the string and θ is the angle between segment XY and axis XP
- The elastic string is in equilibrium, so we can work out the resultant force using:
 - $F = T \cos \theta + T \cos \theta = 2T \cos \theta$
- The $2T$ arises because there is tension on both halves of the string
- The quantity $\cos \theta$ can be found using
 - $\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{30}{50} = \frac{3}{5}$
- So, the resultant force is equal to
 - $F = 2T \cos \theta = 2 \times 4 \times \frac{3}{5} = 4.8 \text{ N}$

5

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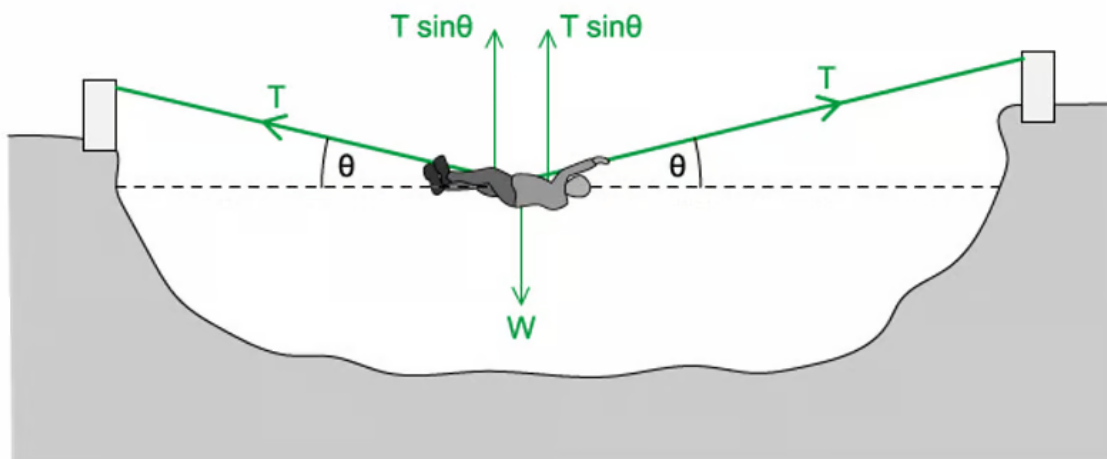


6

The correct answer is **B** because:

- There is a tension T in the rope on each side of the student
- The sum of vertical components of tension should be equal to the weight W

- The vertical component of tension is equal to
 - $T \sin \theta$
- The student is in equilibrium and the vertical forces are shown as in the diagram below, so:
 - $T \sin \theta + T \sin \theta = W$
 - $2T \sin \theta = W$
 - $T = \frac{W}{2 \sin \theta}$

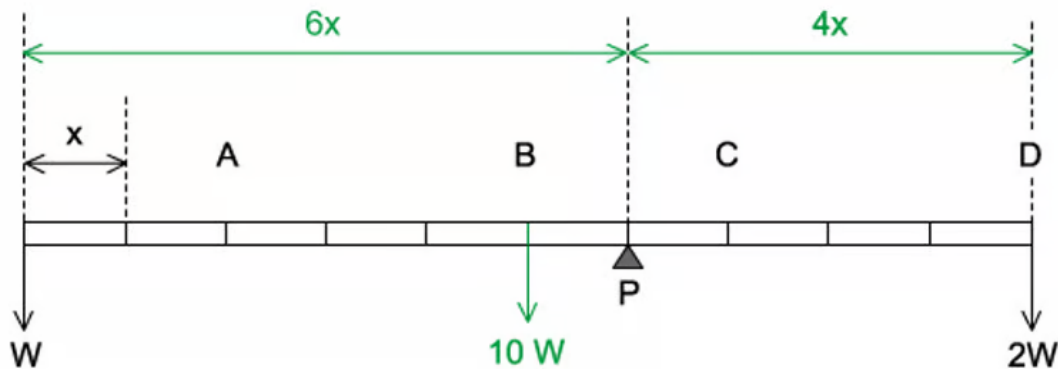


7

The correct answer is **D** because:

- Since the beam is uniform, its weight acts at the centre which is a distance x to the left of the pivot
- For equilibrium : Sum of clockwise moments = Sum of anticlockwise moments
- Clockwise moments = $2W \times 4x = 8Wx$
- Anticlockwise moments = $(W \times 6x) + (10W \times x) = 6Wx + 10Wx = 16Wx$
- Since the anticlockwise moment is greater, the $2W$ weight should contribute to the sum of the clockwise moments, i.e. to the right of the pivot
- The $2W$ weight acts at a distance d from the pivot
- Sum of clockwise moments = Sum of anticlockwise moments
 - $8Wx + 2Wd = 16Wx$
 - $2Wd = 8Wx$
 - $d = 4x$

- A distance of $4x$ to the right corresponds to option D



8

The correct answer is **C** because:

- Since the metre rule is uniform, its weight acts at its centre, which is $5x$
- The weight of the ruler causes a clockwise moment at a distance $5x - 4x = x$ from the pivot
 - Clockwise moment due to the weight of the ruler = $100gx$
- Where g is the acceleration due to gravity
- The weight of the 20 g mass causes an anticlockwise moment at a distance of $10x - 4x = 6x$ from the pivot
 - Anticlockwise moment due to 20 g mass = $20g \text{ mass} = 20g \times 6x = 120gx$
- For equilibrium, the clockwise moment should equal the anticlockwise moment
- Since the anticlockwise moment is greater, the 50 g mass should contribute to the sum of the clockwise moments, i.e. to the right of the pivot
- Let's say the 50 g mass acts at a distance d from the pivot
- Sum of clockwise moments = Sum of anticlockwise moments
 - $100gx + 50gd = 120gx$
 - $50gd = 20gx$
 - $d = \frac{20x}{50} = 0.4x$
- Therefore, the 50g mass should be hung at the $4x + 0.4x = 4.4x$ mark

9

The correct answer is **A** because:

- The beam is uniform, so its weight acts downwards from the centre
- Let's say the weight of the beam is W
- You can choose either point where the cord is attached to the beam as the pivot to get the correct answer, but for this question we will take moments about the left cord (T_1)
- The weight of the beam provides a clockwise moment and the tension T_2 in the cord provides an anticlockwise moment
- Clockwise moment due to the weight of the beam = $\frac{1}{2}LW$
- Anticlockwise moment due to cord $T_2 = \frac{2L}{3}T_2$
- For equilibrium, the clockwise moment should equal the anticlockwise moment
 - $\frac{2L}{3}T_2 = \frac{1}{2}LW$
 - $T_2 = \frac{3}{4}W$
- For equilibrium, the resultant force on the beam should also be zero
- Upward force = Downward force
 - $T_1 + T_2 = W$
 - $T_1 + \frac{3}{4}W = W$
 - $T_1 = \frac{1}{4}W$
- So, the ratio of the two tensions is $\frac{T_1}{T_2} = \frac{W}{4} \div \frac{3W}{4} = \frac{W}{4} \times \frac{4}{3W} = \frac{1}{3}$

10

The correct answer is **A** because:

- So far, we have just looked at situations where the object is uniform so the centre of mass acts through the centre of the object
- We are told the position of its centre of gravity G is a third of the length from the

- $50gd = 20gx$
- $d = \frac{20x}{50} = 0.4x$
- Therefore, the 50g mass should be hung at the $4x + 0.4x = 4.4x$ mark

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10

The correct answer is **A** because:

- So far, we have just looked at situations where the object is uniform so the centre of mass acts through the centre of the object

- We are told the position of its centre of gravity G is a third of the length from the hinge at O so the weight acts at a distance of $\frac{2.7}{3} = 0.9\text{m}$ from O
- For equilibrium, the clockwise moment should equal the anticlockwise moment
- Let's consider moments around the pivot O
- The vertical force F acts at a perpendicular distance of $2.7 \cos 50$
- The weight of the trailer acts at a perpendicular distance of $0.9 \cos 50$
- The vertical force F provides a clockwise moment and the weight of the trailer provides an anticlockwise moment
 - Clockwise moment due to force $F = F \times 2.7 \cos 50$
 - Anticlockwise moment due to the weight = $30000 \times 0.9 \cos 50$
- Sum of clockwise moments = Sum of anticlockwise moments
 - $F \times 2.7 \cos 50 = 30000 \times 0.9 \cos 50$
 - $F = \frac{30000 \times 0.9 \cos 50}{2.7 \cos 50} = 10000 \text{ N}$
- The vertical force is therefore equal to 10 000 N or 10 kN

